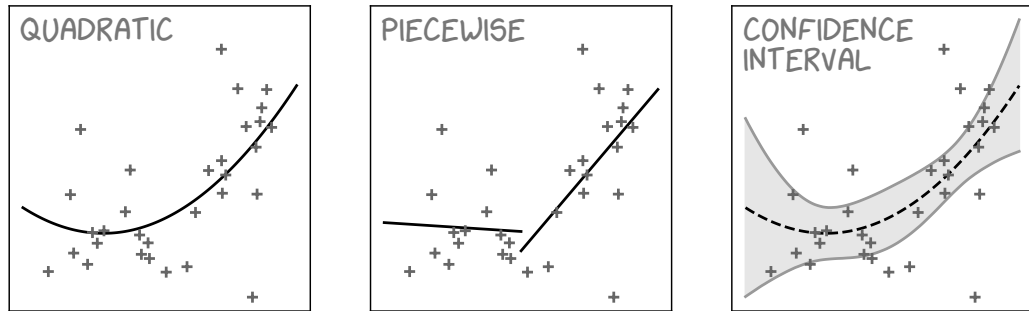


COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 6

5 Data Science (djw1005)



These three plots are adapted from xkcd comic 2048 “Curve-fitting methods and the messages they send” by Randall Munroe (CC BY-NC 2.5)[†].

- | | |
|---------------------|--|
| linear model | (a) Describe a probability model and a fitting procedure that might be behind the “Quadratic” plot. Give pseudocode for the fitting procedure. Explain briefly how the plot is produced. [Note: When describing your model, remember to state any relevant probability distributions.] [7 marks] |
| linear model | (b) Repeat part (a) for the “Piecewise” plot. [5 marks] |
| confidence interval | (c) Repeat part (a) for the “Confidence interval” plot. In your answer, explain carefully the basis for your confidence interval calculation. [8 marks] |

[†] <https://creativecommons.org/licenses/by-nc/2.5/legalcode>

Answer:

— *Solution notes* —

Part (a). Assume we have a dataset (x_i, y_i) , $i = 1, \dots, n$. Consider the probability model

$$Y_i \sim \text{Normal}(\alpha + \beta x_i + \gamma x_i^2, \sigma^2)$$

where α , β , γ , and σ are unknown parameters. We can fit this to the dataset using maximum likelihood estimation; the estimates for the first three parameters correspond to a least squares estimate of the linear model

$$\mathbf{y} \approx \alpha + \beta \mathbf{x} + \gamma \mathbf{x}^2$$

where bold symbols denote vectors. To fit it,

```
model0 = sklearn.linear_model.LinearRegression()
def X0(x): return np.column_stack([x, x**2])
model0.fit(X0(xkcd.x), xkcd.y)
```

The plot shows predicted responses at evenly-spaced values on the horizontal axis, which we can obtain with

```
xnew = np.linspace(0, 10, 100)
ypred = model0.predict(X0(xnew))
```

Marks: 2 for the probability model, 2 for the fitting procedure, 3 for the code including `ypred`. The code here is valid Python, but any pseudocode is acceptable. An answer based on plain maximum likelihood estimation and `scipy.optimize.fmin` is also acceptable.

Part (b). This is similar to part (a), except based on a different linear model:

$$\mathbf{y} \approx \alpha + \beta \mathbf{x} + \gamma \mathbf{1}_{\mathbf{x} \geq 5} + \delta \mathbf{x} \mathbf{1}_{\mathbf{x} \geq 5}.$$

In other words, a linear model with four features. There are two differences in the code. The linear model fit is

```
def X1(x): return np.column_stack([x, np.where(x>=5, 1, 0), np.where(x>=5, 1, 0)*x])
```

and the plotting code shows the response separately over `xnew` in the ranges $[0, 4.999]$ and $[5.001, 10]$, in order to avoid an unsightly jump. *Marks: 2 for the linear model (which most students will probably take a lot more space to write out), 2 for the updated fitting procedure, 1 for the explanation of plotting. Some students are likely to go to a lot of effort to identify coefficients in this model, but smart students will realize they don't have to.*

Part (c). This is based on the quadratic model, but we can find confidence intervals for the curve parameters (α, β, γ) using parametric resampling. First, let $\hat{\alpha}$, $\hat{\beta}$, $\hat{\gamma}$ be the fitted coefficients from part (a). The maximum likelihood estimator for σ is

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{\alpha} + \hat{\beta} x_i + \hat{\gamma} x_i^2 - y_i)^2}.$$

We can parametrically simulate an entire new dataset by

```
\mathbf{Y} = np.random.normal(loc=\mathbf{x}, scale=\hat{\sigma}).
```

We can obtain many estimates for the three curve parameters by repeatedly generating a new dataset and running the fitting procedure from part (a) to it, say $M = 10,000$ times. This gives us M estimates $(\hat{\alpha}_j, \hat{\beta}_j, \hat{\gamma}_j)$, $j = 1, \dots, M$, representing the parameters that a ‘parallel universe’ data scientist might have obtained. We can then plot a 95% confidence ribbon by, for each x in `xnew`, computing M different predictions $\hat{\alpha}_j + \hat{\beta}_j x + \hat{\gamma}_j x^2$, and using

```
lo, hi = np.quantile(predictions, [.025, .975])
```

The confidence ribbon shows $\text{lo}(x)$ and $\text{hi}(x)$, for $x \in \mathbf{xnew}$. *Marks: 3 for resampling the dataset (model and code), 2 for using this to get parameter samples, 3 for turning it into a confidence ribbon. Other methods for finding confidence intervals are also acceptable, e.g. non-parametric resampling (though it's much trickier to get right), and Bayesian.*
